

8th September 2023

Time: 10:30 to 12:00
Max. Marks: 15

06

Q1. Answer the following. (Any Six)

- (1) Malware is a short form of?
- A) Malicious Hardware
 - B) Malicious Software
 - C) Both A and B
 - D) Malicious Script
- (2) Which one option is not a type of cybercrime?
- A) Data Theft
 - B) Installing antivirus for protection
 - C) Damage to data and systems
 - D) Forgery
- (3) How to track Email?
- A) Readnotify
 - B) Self-distracting
 - C) Spy pin
 - D) Gmail
- (4) The purpose of information security is to safeguard:
- A) Confidentiality of data
 - B) Integrity of data
 - C) Availability of data
 - D) All the above
- (5) The primary difference between a virus and a worm is that the host triggers worms, whereas viruses are standard programs that can replicate independently.
- A) True
 - B) False

- (6) Which of the following is/are OWASP's Top 10 vulnerabilities?
- I. Broken Authentication
 - II. Server Memory Leak
 - III. Insecure Deserialization
- A) Both A and B
 B) Both A and C
 C) Both B and C
 D) All of the above
- (7) In which of the following is a person constantly followed/chased by another person or group of several people?
- A) Phishing
 B) Bullying
 C) Stalking
 D) Identity theft
- (8) Which of the following refers to stealing one's idea or invention from others and using it for their benefit?
- A) Piracy
 B) Plagiarism
 C) Intellectual property rights
 D) All of the above
- (9) Tirth is using the 'Cyber Kill Chain' approach to analyze the actions of an intruder on his network. He finds evidence that the most recent activity of the attacker was to successfully use a buffer overflow attack to gain control of a system. What stage is the attacker in?
- A) Attacker in exploitation
 B) Weaponization
 C) Command and Control
 D) Installation
- (10) Which of the following is not a vulnerability-listed site/ database?
- A) Exploit-db
 B) National Vulnerability Database (NVD)
 C) Common Vulnerabilities and Exposures (CVE)
 D) Bing Vulnerability Database (BVD)

Q2. Attempt the following Questions. (Any Three)

09

- (1) Describe any three different cybercrimes.
- (2) Explain any three OWASP vulnerabilities. *XMAS CLOS*
- (3) What is cyber ethics? Describe any three points.
- (4) What is a kill chain in cyber security? Explain any two phases.

DEPARTMENT OF COMPUTER SCIENCE GUJARAT UNIVERSITY
5-year Integrated M.Sc. (Computer Science) SEM V Subject: Cyber Security
Sessional Examination - II

8th December 2023Time: 10:30 to 12:00
Max. Marker: 15

06

Q1. Answer the following. (Any Six)

- (1) What is the default port number for FTP (File Transfer Protocol) in active mode?
A) 20
☒ B) 21
C) 22
D) 23
- (2) Why was Section 66A of the IT Amendment Act, 2008, removed?
A) It was considered outdated and irrelevant in the digital age.
☒ B) It was misused and led to instances of curbing freedom of speech.
C) It conflicted with international cybersecurity standards.
D) It was too lenient in dealing with cybercrimes.
- (3) In cyber forensic investigations, which of the following is a crucial step in preserving digital evidence?
A) Deleting irrelevant files immediately
B) Delaying the investigation to gather more evidence
☒ C) Documenting the chain of custody
D) Ignoring metadata analysis
- (4) What is a common method used in identity theft known as "Phishing"?
☒ A) Sending fraudulent emails to trick individuals into revealing personal information
B) Physical break-ins
C) Stealing physical documents
☒ D) Interception of postal mail
- (5) What is the primary role of cyber ethics in the digital realm?
A) Facilitating unrestricted access to information
B) Promoting hacking for educational purposes
☒ C) Encouraging anonymity in online activities
☒ D) Guiding responsible and ethical behavior in cyberspace
- (6) How does blockchain contribute to cybersecurity?
A) By centralizing control over sensitive information
B) By enhancing data confidentiality
☒ C) By reducing the need for encryption
D) By relying on a single point of failure
- (7) What is the primary purpose of a digital signature in the context of cybersecurity?
☒ A) To encrypt confidential messages
B) To establish a secure connection between devices
☒ C) To verify the identity of the sender
D) To prevent unauthorized access to a network
- (8) How does a firewall enhance cybersecurity?
A) By encrypting data during transmission
☒ B) By preventing unauthorized access and controlling incoming/outgoing network traffic
C) By detecting and removing malware from devices
D) By automatically updating software to patch vulnerabilities

Authorized

- C) By detecting and removing malware from devices
- D) By automatically updating software to patch vulnerabilities

- (9) What is the primary objective of cybercriminals?
 - A) Spreading malware, spam and viruses
 - B) Promoting online safety
 - C) Enhancing cybersecurity measures
 - D) Collaborating with law enforcement agencies
- (10) What is a common method used in digital payment frauds?
 - A) Cash transactions
 - B) Two-factor authentication
 - C) Phishing attacks
 - D) Secure SSL connections

Q2. Attempt the following Questions. (Any Three)

- (1) What is cryptography? How is it useful in the context of cybersecurity?
- (2) What is the difference between a firewall, IDS and IPS?
- (3) What is the chain of custody in digital forensics? Explain in detail.
- (4) What exactly is personal cybersecurity? What is it aimed at protecting?

Time : 3-00 Hours]

Total Marks: 35

Instructions:

- Draw Diagrams wherever necessary.

Q-1 Answer the following. (Any Eleven)

11

1. Describe Cyber Crime.
2. What exactly is identity theft?
3. Give four examples of cybercrime against the government.
4. What is a vulnerability?
5. Describe the cyber kill chain and names of the phases.
6. Write a full form of OWASP.
7. What is Cryptography?
8. Write a definition of Darkweb.
9. What exactly is Blockchain?
10. Confidentiality, Integrity, and Availability form what is often called the _____.
11. A symmetric encryption scheme has five ingredients: plaintext, encryption algorithm, ciphertext, decryption algorithm and _____.
12. Child pornography is an offence under section _____.
13. Port number _____ is used for PoP3.

Q-2 Attempt the following:

12

1. Explain the differences between threats and vulnerabilities.
2. Write down any three forms of cybercrime.

OR**Q-2 Attempt the following:**

1. Explain Cyber Kill Chain in details.
2. Brief National Cyber Security Policy of India.

Q-3 Attempt the following:

12

1. Explain any three vulnerabilities from OWASP's top 10.
2. What is the chain of custody in digital forensics?

OR**Q-3 Attempt the following:**

1. Brief any three social media crimes along with its cyber law.
2. What is Common Vulnerabilities and Exposures? explain.

DEPARTMENT OF COMPUTER SCIENCE
GUJARAT UNIVERSITY
5-year Integrated M.Sc.(Computer Science)
SEM V

Subject: Java Programming
Sessional Examination – I

5th September 2023

Time: 1 hr 30 min
Max. Marks: 30

Q1. Answer the followings. (Any Nine)

18

1. Explain the use of 'final' keyword.
2. Enlist access specifiers in Java.
3. Differentiate between break and continue statement.
4. What is the use of super keyword?
5. Why is Java known as platform independent?
6. Differentiate between abstract class and interface.
7. What is Copy Constructor.
8. What is the difference between JDK, JRE, and JVM? 1.0
9. Difference between method overloading and method overriding. 1.5
10. What is accessor and mutator?

Q2. Attempt the following Questions. (Any Three)

12

1. What is the constructor? Explain the types of constructors are used in Java with constructor overloading. 1.7
2. Difference between C++ and Java. 1.0
3. How we can achieve multiple inheritance in Java, explain with example. 1.1
4. Explain different types of polymorphism with example.
5. Explain different uses of this keyword in Java. 1.6

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BEST OF LUCK

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Department of Computer Science
GUJARAT UNIVERSITY
5 Year Integrated M. Sc. (Computer Science)
Semester : V
Subject Name : Java Programming

Time : 1.5 Hours
Date: 4th Dec, 2023

Total Marks : 30

Q-1 Attempt the following

1. Explain enum keyword and provide example. 2.28
2. Explain finalize() method with an appropriate example. 2.18
3. What is a native method in Java, and when is it used? 2.19
4. What is casting in Java? Discuss implicit and explicit casting. 2.27
5. Difference between method overloading and method overriding. 2.13 4

[10] / 6

Q-2 Attempt the following (Any 2)

1. Discuss the purpose of the try, catch, and finally blocks in exception handling. 3.3
2. Explain the difference between a process and a thread. Discuss the advantages of using multithreading in a program. 3.13
3. What is the purpose of the RandomAccessFile class? Explain how you can perform both read and write operations on a file using RandomAccessFile. 3.11
4. How does the wait and notify mechanism work in Java threading? Provide an example of using wait and notify for inter-thread communication. 3.17

[10] / 5

Q-3 [A] Attempt the following (True / False):

1. The List interface in Java represents an ordered collection that allows duplicate elements.
2. Duration class represents a time span, such as "2 hours and 30 minutes."
3. The Math.random() method is part of the Random class in Java.
4. The DateTimeFormatter class in Java can be used to parse and format dates and times.
5. Regular Expressions are patterns used for matching character combinations in strings.

[10] / 5

[B] Attempt the following (Any 2)

1. Explain the significance of OffsetDate and ZonedDateTime in the java.time package. 4.2
2. Explain the concept of lambda expressions with example. 4.13
3. Explain how to generate random numbers in Java.

Integ. MSc (CS) Sem.-5 Examination
Java Programming
December-2023

Time : 3-00 Hours]

[Max. Marks : 70

Instructions:

- Write both the Sections in the separate answer book.
- Both Sections having equal weightage.
- Draw Diagrams wherever necessary.
- Make Assumptions wherever necessary.

SECTION – I

Q-1 Explain the following terms in one or two line only: 07

1. A keyword that refers to parent class member is _____.
2. Operators like >, <, !=, == are called as _____.
3. _____ method checks whether entered character is Digit or not.
4. Which Access Modifier is visible in non-child class of other package?
5. What is nextLine() method?
6. The interface is used to implement the _____ concept of java.
7. Why we use finally?

Q-2 Attempt the following: 14

1. Explain static, final and super keywords by giving a suitable example.
2. Explain the concept of lambda expressions with example.

Q-2 **OR**

Attempt the following:

- ✓1. Explain the difference between a process and a thread. Discuss the advantages of using multithreading in a program.
- ✓2. Discuss the purpose of the try, catch, and finally blocks in exception handling.

Q-3 Attempt the following: 14

1. What is the Inheritance? Explain types of inheritance in java.
2. Define polymorphism in Java and provide examples of its usage. Explain the difference between compile-time and runtime polymorphism

OR

Q-3 Attempt the following:

- ✓1. What is the constructor? Explain the types of constructors are used in Java with constructor overloading.
- ✓2. Define enumeration types in Java and Discuss the enum keyword and provide examples.

P.T.O

861-2
SECTION - II

Q-4 State True/False with proper Justification. 07

- ☒ 1. instanceof is not an operator in java.
- ☒ 2. We cannot create custom exception by extending the Throwable class. ☒
- ☒ 3. main() method can be declared final. ☒
- ☒ 4. static variable can be used in non-static methods. ☒
- ☒ 5. public is the default access specifier for the members of class in java. ☒
- ☒ 6. abstract class cannot have constructors.
- ☒ 7. The Math.random() method is part of the Random class in Java. ☒

Q-5 Attempt the following: 14

1. Differentiate between abstract class v/s interface.
2. How does the wait and notify mechanism work in Java threading? Provide an example of using wait and notify for inter-thread communication.

OR

Q-5 Attempt the following:

- ☒ 3. Differentiate between method Overloading v/s method Overriding.
- ☒ 4. Explain the significance of OffsetDate and ZonedDateTime in the java.time package.

Q-6 Attempt the following: 14

- ☒ 1. Describe the characteristics and use cases of the Queue interface.
- ☒ 2. Compare and contrast List and Set interfaces in the Java Collection Framework.

Q-6 **OR**

Attempt the following:

1. Explain the role of regular expressions in Java. Provide examples of using regular expressions for string manipulation and pattern matching..
2. Differentiate between InputStream and OutputStream. Explain the purpose of buffered streams in Java I/O.

—X—

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Subject: Operating Systems
Date: 04/09/2023

Time: 1 hr 30 min
Max. Marks: 30

Q1. Define following terms:

1. Interrupt
2. Starvation and Aging 2.2d
3. Role of linker and loader 1.10
4. Context Switch
5. Program Counter in PCB

Q2. Answer the followings: (Any Five)

- 1.8 What is process? Explain IPC (Inter Process Communication) in detail.
- 1.3 What are the different services provided by operating system? Give brief detail for each. 9
- 3 Mention the Scheduling Criteria? Brief about each.
- 2.3 What is the Critical-Section Problem? What is semaphore? Explain in detail.
- 1.4 Write a note on System calls. C
- 2.5 What is Deadlock? Which are the conditions in which deadlock can arise?

Q3. Process and its corresponding burst time and arrival time is given. Mention the Gantt Chart and Find the average turnaround time and average waiting time for FCFS, SJF (Preemptive and Non-preemptive), Round Robin (Quantum Time = 2ms) and Priority based (Non-preemptive) algorithm. Also mention which algorithm will be best.

Process	Arrival Time	Burst Time	Priority
P1	0	7	3
P2	2	3	2
P3	4	2	1
P4	5	5	4

$$CTD - AT = TAT$$

$$TAT - BT = WT$$

P1 - 5 3
P2 - 1 2
P3 - 5 4

RR

Q4. Write a bankers/Safety algorithm and answer the question given below:
Consider the following data for the system for five processes and four resources.

Need =	Allocation				Max	Available						
	A	B	C	D		A	B	C	D			
P0	0	0	1	2	0	0	2	2	1	5	2	0
P1	1	0	0	0	1	7	5	0	1	5	3	2
P2	1	3	5	4	2	3	5	6	2	8	6	0
P3	0	6	3	2	0	6	5	2	2	14	11	8
P4	0	0	1	4	0	6	5	6	2	14	13	12
					3				0	6	4	2

- Calculate the matrix need.
- What will be the safe sequence?
- Is the system being in safe state?
- If P1 arrives with request {0, 3, 3, 0} can it be granted immediately at initial stage?

3, 14, 12, 12

1.40
-1.25

Department of Computer Science
5 Year Integrated M.Sc. (Computer Science) Sem - 5
Operating Systems
Sessional 2 Examination
December 2023

Time: 90 mins

Total Marks : 30

Notes:

1. Write side figure indicates marks.
2. Answer any THREE from Q2 to Q5.
3. Write to the point answer.

- Q1. Answer following: (Any Five) 10
- a. What is fragmentation? 3.2
 - b. Explain Thrashing in brief. 3.7
 - c. What is belady's anomaly? 3.8
 - d. Why demand paging required? 3.6
 - e. What is link? Differentiate between hard link and soft link.
 - f. What is role of memory management unit?
- Q2. What is paging? Briefly explain address translation scheme and hardware of paging. 4
- Q3. What is the need of RAID? Explain different RAID levels. 3.15 4
- Q4. Explain any two text processing commands. 4
- Q5. List out File Allocation Methods and explain any two in detail. 3.22 4
Index Atomic, FAT, 3e
- Q6. Estimate the Number of page faults for the following page reference string for frame size 3. 4
7, 0, 1, 2, 0, 3, 4, 0, 2, 3, 0, 1, 2, 0, 1, 7, 0, 1, 3
Use FIFO, Optimal and LRU Algorithms.
Which Algorithm performs better?
- Q7. A disk system has 75(0-74) cylinders, the read/write head is at cylinder 37 and determine the order of head movement for FCFS, SSTF and C-SCAN to satisfy the following stream of request. Also answer followings: 4
1. Total head travel distance
 2. What is the time taken to satisfy all the requests if it takes 0.7 milliseconds to move from one cylinder to an adjacent one?
 3. Which algorithm is better?

Disk request: 12, 20, 25, 70, 41, 61, 38, 40, 63

12, 20, 25, 38, 40, 41, 61, 63, 70

Integ. MSc (CS) Sem.-5 Examination
Operationg Systems
December-2023

Time : 3-00 Hours]

[Max. Marks : 70

Instructions:

- Write both the Sections in the separate answer book.
- Both Sections having equal weightage.
- Draw Diagrams wherever necessary.
- Make Assumptions wherever necessary.

SECTION - I

Q-1 Define following terms:.

11

- ✓ 1) Interrupt
- ✓ 2) Starvation
- ✓ 3) External fragmentation
- ✓ 4) Role of linker and loader
- ✓ 5) Context Switch
- ✓ 6) Lazy swapper and pager
- ✓ 7) Thrashing
- ✓ 8) Program Counter in PCB
- ✓ 9) Demand paging
- ✓ 10) In command
- 11) BIOS

Q-2 Attempt the following. (Any Three)

12

- ✓ 1) What is process? Explain IPC (Inter Process Communication) in detail.
- ✓ 2) Mention the Scheduling Criteria? Brief about each.
- ✓ 3) What is the process control block? What are the information store in process control block.
- ✓ 4) What is the Critical-Section Problem? What is semaphore? Explain in detail.

Q-3 Attempt the following. (Any Three)

12

- ✓ 1) Write Banker's algorithm and explain.
- ✓ 2) What is system call? Explain how system call works? Also Mention few system calls.
- ✓ 3) Write six steps for Direct Memory Access and explain.
- ✓ 4) Explain following :
 - a. The Role of the Linker and Loader
 - b. List different attributes of file and brief about each.

Size name

(P.T.O)

SECTION - II

Q-4 Answer the following. (Any Two)

08

- 1) Write a note on Paging Hardware With TLB.
- 2) Brief about following commands:
ls, chmod, ps, wc
- 3) Explain unix and windows Operating System Structure.

Q-5 Attempt the following.

15

1. Process and its corresponding burst time and arrival time is given. Mention the Gantt Chart and Find the average turnaround time and average waiting time for **FCFS, SJF(Preemptive and Non-preemptive), and Round Robin(Quantum Time = 2 ms)** algorithm. Also mention which algorithm will be best.

Process	Arrival Time	Burst Time
P1	0	7
P2	2	3
P3	4	3
P4	5	4

2. A disk system has 80(0-79) cylinders, the read/write head is at cylinder 45 and determine the order of head movement for FCFS, SSTF, SCAN and C-SCAN to satisfy the following stream of request. Also answer followings:
 - a) Total head travel distance
 - b) What is the time taken to satisfy all the requests if it takes 1 milliseconds to move from one cylinder to an adjacent one?
 - c) Which algorithm is better?

Disk request : 60, 45, 70, 32, 40, 8, 78, 65, 10.

3. Estimate the Number of page faults and page hits for the following page reference string for frame size 3:
7, 0, 1, 2, 0, 3, 2, 7, 2, 1, 0, 5, 0, 7
Use FIFO, Optimal and LRU Algorithms. Also mention which algorithm is better.

Q-6 Attempt the following. (Any Two)

12

- 1) Explain Text processing commands with different options in Linux.
- 2) What is the need of RAID? Explain different RAID levels.
- 3) List out File Allocation Methods and explain each in detail.

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DEPARTMENT OF COMPUTER SCIENCE
GUJARAT UNIVERSITY
INTEGRATED MSc (COMPUTER SCIENCE) – SEM – V
SUBJECT: CRITICAL THINKING (THEORY)

TIME: 60 MINS

Q1

MAX MARKS: 15

[15 marks]

Answer any 5 in detail:

- ✓ 1. Define: Claims, Give example of atleast 2 claim statements
- ✓ 2. What do you understand by an "argument"? Give atleast 2 examples of an argument.
- ✓ 3. State name of any 3 western philosophers and give their contributions in brief *Aristotle, Plato, Socrates*
4. State name of any 3 indian philosophers and give their contributions in brief.
- ✓ 5. State the benefit of studying "Critical Thinking" *Patangali - Yoga*
- ✓ 6. State whether the following are valid claims or not :
 - a. $15 * 3 = 45$ ✓
 - b. "Kitne Aadmi They Kalia?" ✗
 - c. Bill Gates Founded Apple Corporation ✓

sports sports supp

DEPARTMENT OF COMPUTER SCIENCE
GUJARAT UNIVERSITY
M.Sc (CS), SEMESTER - 5
SUBJECT: CRITICAL THINKING (Theory)

Max Marks:15

Time:60 Minutes

Q1 Answer the followings in 2-3 sentences: (Any 1)

[03
marks]

1) Give any 2 examples of deductive arguments and justify why they are deductive arguments

2) Discuss following in brief:

(a) Deductive Arguments (b) Modus Ponens (c) Modus Tollens

*If p then q
not q therefore not p*
If premises true, conclusion must be true
the mode of putting put p, get q
(If p then q, p, therefore q)

Q2 Do as Directed:

[12
marks]

1) Phenomenon : "Sunlight affects plant growth"

For the above phenomenon, formulate the followings:

- (a) Hypothesis
- (b) Independent Variable
- (c) Dependent Variable
- (d) Explanation
- (e) Prediction
- (f) Evaluation

2) Question: "Which food is healthier", you are given two choices:

- (i) Fast food (eg. Pizza)
- (ii) Boiled food (eg. Khichdi)

For the above question, formulate the followings for each choice:

- (a) Claim
- (b) Evidence
- (c) Reasoning

Integ. MSc (CS) Sem.-5 Examination
Critical Thinking
December-2023

Time : 3-00 Hours]**[Max. Marks : 35****Q1 Answer the followings in 2-3 sentences: (Any 5)****[10 marks]**

- 1) What is an argument? Give an example.
- 2) How claims are different from arguments?
- 3) Differentiate between dependent variable and independent variable
- 4) Do you think experiments are always needed to support your claims?
- 5) Give atleast 2 examples of evidences.
- 6) Give any 2 examples of deductive arguments and justify why they are deductive arguments
- 7) What is a claim? Give an example of claim that is valid and another example that is invalid

Q2 Do as Directed: (Any 2)**[12 marks]**

- 1) What is Critical Thinking? State the benefits of critical thinking and how does it change our personality?
- 2) Phenomenon : "Sunlight helps in synthesizing Vitamin D"
For the above phenomenon, formulate the followings:
 - (a) Hypothesis
 - (b) Independent Variable
 - (c) Dependent Variable
 - (d) Explanation
 - (e) Prediction
 - (f) Evaluation
- 3) Question: "Which water is safe for drinking?", you are given two choices:
 - (i) RO Purified Water
 - (ii) Tap Water

For the above question, formulate the followings for each choice:

- (a) Claim
- (b) Evidence
- (c) Reasoning

Q3 Do as directed (Any 2)**[08 marks]**

- 1) State whether the following statements are valid claims or not? Give appropriate reasons:
 - a) Delhi is the capital of India
 - b) Does photosynthesis require sunlight?
 - c) $5 * 2 = 10$
 - d) Good Morning
- 2) Discuss the contribution of any 4 critical thinkers/philosophers
- 3) Write a short note on application of Critical Thinking Skills in the area of Computer Science

Q4 What is 'Deductive Argument'? Discuss Modus Ponens and Modus Tollens giving examples of each.

[05 marks]

DEPARTMENT OF COMPUTER SCIENCE
GUJARAT UNIVERSITY
M.Sc (CS), SEMESTER - 5
SUBJECT: ADVANCED NETWORKING (Theory)

Max Marks:30

Q1 Answer the followings in 2-3 sentences: (Any 6)

- ~~9)~~ Discuss the AIMD principle in brief
- ~~10)~~ What is the use of SYN and PSH flags in TCP?
- ~~11)~~ What is the use of CRC? *Cyclic Redundancy Check*
- ~~12)~~ What is the role of 3-way handshake.
- 13) What is SDN? *Soft Define Net*
- 14) What is the use of longest - prefix match?
- ~~15)~~ How RTT is helpful to increase the throughput of Internet?
- ~~16)~~ What is the role of Network Layer?

Q2 Do as Directed: (Any 3)

- ~~6)~~ Differentiate between TCP and UDP, also explain how python programming differs in TCP & UDP
- ~~7)~~ What is error correction & error detection. Discuss parity techniques
- ~~8)~~ Write a short note on Internal of Router.
- 9) Write a short note on fragmentation used in IP address
- 10) What is the role of optional field in IP packet? Discuss the use of any 2 optional fields.

*Parity Bit
check sum*

*ACK
SYN
PSH
FIN*

TIME: 90 MINS

MAX MARKS: 30

Q1

Briefly explain the use of following Protocols: (Any 5)

[10 marks]

1. SMTP
2. HTTP
3. DNS
4. IP
5. MIME Base 64
6. IMAP

Q2

Answer any 4 in detail:

[20 marks]

1. Discuss any 4 layers of OSI Model in detail.
2. Discuss how Email protocols work.
3. Write a short note on working of DNS
4. What is the difference between Wired and Wireless Transmission Media? Discuss Wired Transmission Media, its types and explain each in short.
5. What is Delay? Discuss the types of delay in brief.
6. Answer any two of the followings:
 - a. Time Division & Frequency Division Multiplexing
 - b. Nyquist Criteria & Shannon's Capacity
 - c. Client-Server Model & Peer to Peer Model

Max Marks:30

Q1 Answer the followings in 2-3 sentences: (Any 6)

- ~~Q1~~ Discuss the AIMD principle in brief
- ~~10)~~ What is the use of SYN and RSH flags in TCP?
- ~~11)~~ What is the use of CRC? *-Cyclic Redundancy Check*
- ~~12)~~ What is the role of 3-way handshake.
- 13) What is SDN? *Soft Define Net*
- 14) What is the use of longest - prefix match?
- ~~15)~~ How RTT is helpful to increase the throughput of Internet?
- ~~16)~~ What is the role of Network Layer?

Q2 Do as Directed: (Any 3)

- ~~5)~~ Differentiate between TCP and UDP, also explain how python programming differs in TCP & UDP
- ~~7)~~ What is error correction & error detection. Discuss parity techniques
- ~~8)~~ Write a short note on Internal of Router.
- 9) Write a short note on fragmentation used in IP address
- 10) What is the role of optional field in IP packet? Discuss the use of any 2 optional fields.

*Parity Bit
check sum*

*ACK
SYN
PSH
FIN*

Integ. MSc (CS) Sem.-5 Examination
Advanced Networking
December-2023

Time : 3-00 Hours]

[Max. Marks : 70

- Instructions: 1] Figure to the right indicates full marks.
2] All Questions are compulsory.

SECTION - 1

Q1 Answer the following in brief:

[20 marks]

- 1) How many hosts can be supported by a single class 'B' address?
- 2) What is the purpose (use) of Loopback address?
- 3) Is IP header of fixed size or is it variable size? Justify your answer.
- 4) Give two differences between wired & wireless networking
- 5) Justify the following statement: "Top layer protocols are hardware independent"
- 6) State in brief, the role of Internet Society in networking.
- 7) State the command to check network connectivity between two computers.
- 8) State the size of field used to identify port number in UDP/TCP header.
- 9) What is topology? Which topology requires a device 'switch'?
- 10) State the port number of HTTP & SMTP protocol.

Q2 Do as directed

[10 marks]

- 1) Write a short note on various delays in network communications.
- 2) Why does computer networking require two different addresses? What is the role of hardware address (physical address) and how does it differ from IP address?

OR

- 1) Explain the concept of packet fragmentation in IP. Why does it take place?
- ✓ 2) State the use of following commands:
 - a) ipconfig
 - b) ping
 - c) nslookup
 - d) arp
 - e) route

Q3 Do as directed (Any one):

[05 marks]

- 1) Differentiate between the following:
 - a) UDP and TCP
- 2) What is the significance of round trip time (RTT)? Explain how TCP calculates RTT. Justify the statement: "An optimum RTT value estimation leads to better throughput".

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SECTION - 2

Q4

Answer the following in brief:

[20 marks]

- 1) Explain the significance of strict source route option in IP.
- 2) State the usage of TELNET.
- 3) Why do we need DNS?
- 4) Explain the term – "Timeout", give an example.
- 5) When do we use VPN & NAT.
- 6) State the use of FIN flag in TCP.
- 7) Discuss the need of TTL field in packets.
- 8) Discuss the concept of forwarding & caching in HTTP.
- 9) Give the usage of MIME.
- 10) Explain forwarding & routing with reference to routers

Q5

Do as directed

[10 marks]

- ✓ 1) What is tail-drop? Explain the significance of RED with TCP and how does it improve on tail-drop?
- 2) Discuss AIMD in detail.

OR

- 1) Write a short note on Wired Transmission Media.
- 2) Discuss the protocols of E-mail in detail

Q6

Do as directed (Any one)

[05 marks]

- 1) What is error correction & error detection?
- 2) State the functionality of Network Layer & Transport Layer.